Gonzalo Díaz López

Software Engineer (C++, Python) · PhD in Physics

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Summary

Software engineer (**C++**, **Python**) with a PhD in physics and eight years of building scientific software. Main developer of a public data-processing framework at CNRS, including its packaging and **continuous integration**; earlier rewrote an experiment's core software, now the base other developers build on. Adapts quickly to new technical domains across three scientific projects, and has taught at university.

Experience

Software Research Engineer

CNRS, LPNHE / Sorbonne Université, Paris — *Apr 2025 – Present*

- Lead development of **Allen Core**, a public **C++** framework for processing scientific data on **CPUs and GPUs**, derived from the real-time system of CERN's LHCb experiment (gitlab.cern.ch/allen-core).
- Build a demonstrator that proves the framework is generic and usable by other experiments: it runs another project's complete algorithm chain, on either GPU or CPU.
- Package the framework for others to adopt: installable build, **containers** and **continuous integration**.

Postdoctoral Researcher

CNRS, LPNHE / Sorbonne Université, Paris — *Apr 2023 – Mar 2025*

- Rewrote, single-handedly, Hyper-Kamiokande's software that turns detector signals into physics events, from deprecated **C++** to **Python**: faster, easier to debug, and now the base for PhD students' projects.
- Ran simulation productions on **high-performance computing (HPC)** clusters to test and validate the new software.
- Supervised a Master's student through a six-month project.

PhD Researcher

IGFAE, Universidade de Santiago de Compostela — *Sep 2018 – Feb 2023*

- Made the NEXT-100 detector simulation 1,000× faster by replacing repeated calculations with a lookup table, so a dataset equal to 20 years of detector operation became feasible.
- Implemented that simulation in **C++** and contributed **Python** software for data reconstruction and analysis.
- Implemented **machine-learning** methods that improved how detector events are classified.

Skills

- **Programming:** C++, Python, Bash, CUDA (basic)
- **Engineering:** object-oriented design, multi-threading, memory debugging (gdb), code review
- **Build & delivery:** Git, CMake, GitLab CI/CD, Apptainer, Docker, pytest
- **Computing:** Linux, HPC clusters (Slurm), GPU computing
- **Data & machine learning:** NumPy, Pandas, HDF5, Matplotlib, scikit-learn, PyTorch, Keras

Education

PhD in Particle Physics
Universidade de Santiago de Compostela *2018 – 2023*

Thesis: *Sensitivity of the NEXT-100 detector to the search of neutrinoless double beta decay*.

MSc in Nuclear and Particle Physics
Universidade de Santiago de Compostela *2017 – 2018*

Thesis: *Double beta identification studies in the NEW detector with neural networks*.

BSc in Physics
Universidade de Santiago de Compostela *2013 – 2017*

Thesis: *Critical Phenomena and Gravitational Collapse*.

Teaching

- **Numerical Physics**, Master's level (36 h) — Université Paris Cité, 2024–25.
- **Multivariate analysis** (Mathematical Methods III) and **physics lab sessions**, Bachelor's level (20 h/year each) — Universidade de Santiago de Compostela, 2019–21.

Publications

- *Energy calibration of the NEXT-White detector with 1% resolution near Qββ of ^{136}Xe*, JHEP 10 (2019) 230 (co-author).
- *Demonstration of the event identification capabilities of the NEXT-White detector*, JHEP 10 (2019) 052 (co-author).

Languages

Spanish (native) · Galician (native) · English (professional) · French (intermediate)